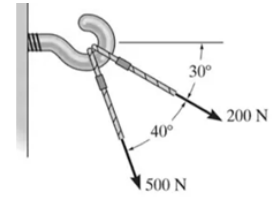


1. Determine the magnitude and resultant of resultant force acting in the hook

Ans: $R = 667\text{ N}$

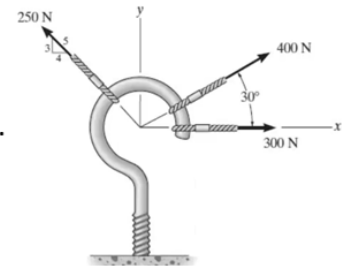


2. Five forces 10N, 20N, 30N, 40N and 50 N are acting at point such that angle between them are equal. If 10N force is acting in the positive X-direction in horizontal direction , find out the resultant force of the system.

Ans: $R = 42\text{ N}$, angle = 234°

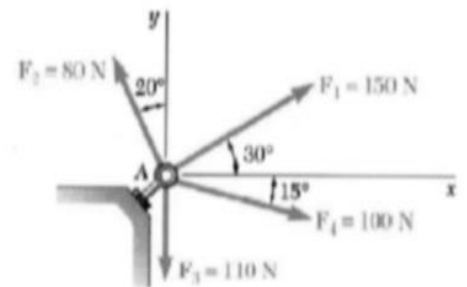
3. The three forces acting on the screw as shown in figure . compute the magnitude and direction of the resultant force.

Ans: $R = 567\text{ N}$, angle = 38°



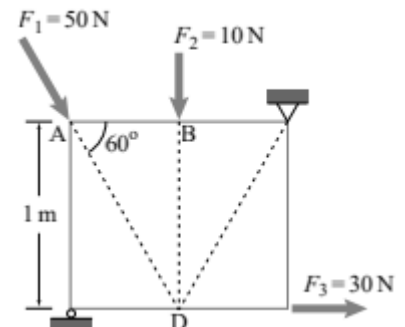
4. . Forces F_1 , F_2 , F_3 , and F_4 are acting at point A. Determine magnitude and direction their resultant by graphical method

Ans: $R = 200\text{ N}$ Angle: 4° from $+x$



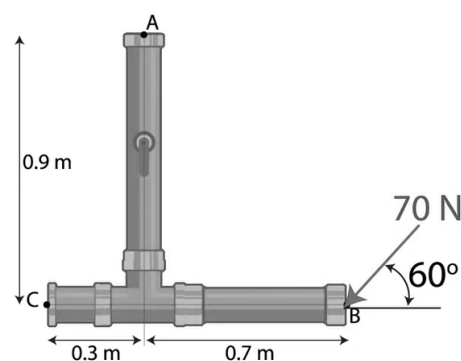
5. In the figure shown, $F_1 = 50\text{ N}$, $F_2 = 10\text{ N}$, $F_3 = 30\text{ N}$, and $\theta = 60^\circ$. Find the equivalent force-couple system about point D of the structure.

Ans: $F_c = 82\text{ N}$, $M_A = 74\text{ Nm}$

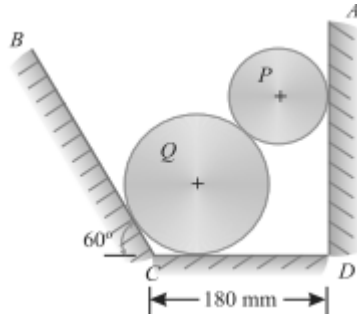


- 6.

The 70-N force acts on the end of the pipe at B. Determine (a) the moment of this force about point A, and (b) the magnitude and direction of a horizontal force, applied at C, which produces the same moment.

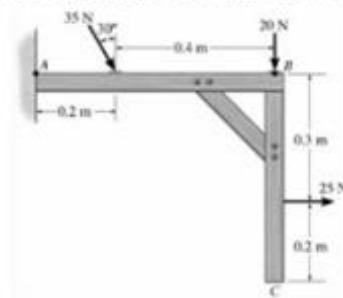


7. Two cylinders P and Q rest in a channel as shown in Fig.. The cylinder P has diameter of 100 mm and weighs 200 N, whereas the cylinder Q has diameter of 180 mm and weighs 500 N. Determine the force acting on all the sphere. Ans: 134N, 241N



8.

A 2-D force and couple system as shown in fig below. Find the equivalent resultant force and couple moment acting at A and then the equivalent single force location along the beam AB.



Ans : $R = 66 \text{ kN @ } 50^\circ$ in 4th quadrant
 $M = 105.6 \text{ kNm}$
 Dist = 2.1m from A

9. An electric light fixture weighing 15 N hangs from a point C, by two strings AC and BC. The string AC is inclined at 60° to the horizontal and BC at 45° to the horizontal as shown in Fig

